Browser-Based Game for Teaching Cyber Security Concepts

Project Plan

Elizabeth Eardley

ISG Final Year Project

Supervised by: Dr Rachel Player
Royal Holloway, University of London

1 - Abstract

Game-based learning combines entertainment and education to create immersive and interactive experiences that promote active learning. Recent studies highlight that educational games have become more reckognised as effective learning tools across many disciplines.

In a study conducted by Begg, Dewhurst and Macleod (2005), two narrative-based scenarios from the University of Edinburgh's medical curriculum demonstrated how problem-based learning principles can transform traditional instruction. In these activities, students respond to evolving situations presented in an engaging, game-like style uncommon in standard university coursework. The study found that successful game-informed learning followed several key principles. First, the game needed a compelling backstory and context to provide emotional entry points and relevance for the learner. Continuous feedback ensured that learning felt rewarding and progress was visible. Learners were also given opportunities for safe failure, allowing them to make mistakes and try again without real-world consequences. Players adopted meaningful roles that shaped their behavior and engagement within the learning environment, while emotional connections made the experience more memorable and authentic. Finally, consequential decision-making—where players' actions directly influenced outcomes—helped create a strong sense of motivation and active participation in the learning process.

Similarly, James Paul Gee (2004) argued that games could be powerful learning systems, which enable learnes to "learn by doing" through active co-creation of their experiences. Video games, in general, combine motivation, challenge and feedback in ways that would make learning more fun and engaging. Additionally, games also incourage exploration, experimentation and productive failure establishing a very healthy relationship to continuos learning and improvement.

From a broader perspective, de Freitas (2018) observed that research on the use of games as educational tools is fragmented across many disciplines such as education science, game studies, neuroscience and information science - resulting in inconsistent terminology and methodology. She also identified the ongoing challenge of aligning fun and playability with learning outcomes, putting an emphasis on the need for thoughtful design and planning that balances fun and education. De Freitas also argued that many studies highlight that game based learning show positive cognitive and behavioural impacts to the learner, improving knowledge retention, motivation and skill development.

Building further on these theories of learning, the Dreyfus Model of Skill Acquisition (1980) offers insight into the gradual development of expertise over time. The model proposes five stages of skill acquisition where each stages represents a deeper understanding on the subject being learnt. Applying this model within the game would ensure scaffolding learning, where players can build confidence, competence and evolve their abilities over time.

This project proposes the development of a browser-based, story-driven game to teach cyber security concepts through quest-like challenges. Each quest will introduce a topic - ranging from understanding what cyber security is and common risks/attacks to more advanced areas such as access control and cryptography. By including a narrative, continuos feedback and skill progression through the quests, the game aims to transform cybersecurity education into an engaging and meaningful learning experience accessible to anyone.

2 - Timeline

By the end of Term 1, the aim is to complete a basic prototype including all quests and a simple final level to put the learners knowledge to the test. In Term 2, development will continue with the integration of a learning loop and player progress tracking. The Term 1 timeline is presented below:

Note: Week 11 has been allocated as a buffer week to accommodate any unforeseen delays in development or testing.

WEEK	DELIVERABLES
1-2	Research and review serious games for cybersecurity education and related learning strategies. Design a document outlining game mechanics, educational objectives and user flow including wireframes
3	Familiarise with Phaser and polish javascript and typescript knowledge. Set up project repository with all requirements
4-5	Find/create character animations and background art for the game. Create the welcome page and the quests introducing: ● What is cybersecurity ● Understanding cyber risk Work on the report/documentation
6-7	Create the quests introducing: ● Common cyber threats ● The CIA triad ● IAAA ● People and security Work on the report/documentation
8-9	Create the quests introducing: ● Firewalls, antivirus, encryption, IDS ● Operating systems ● Controls (technical, administrative and physical) ● Cryptography basics Work on the report/documentation
10	Create a simple final boss level including all topics Work on the report/documentation

3 - Risks and Mitigations

3.1 Poor estimation of tasks

Inaccurate estimation of time for each deliverable could result in missed deadlines or incomplete deliverables. To mitigate this, I have allocated a buffer week in the occasion of unforeseen delays, tasks will be broken down to smaller and clearly defined milestones with specific deadlines and make use of project management tools to visualise progress. I will also continuously review progress and adjust the timeline as needed.

3.2 Hardware failure

Unexpected hardware or software failure could lead to delays and/or data loss. To mitigate this, I plan to maintain regular backups of the project and any documentation onto the version control system provided by the university.

3.3 Uneven balance between coding and documentation

Focusing too heavily on coding could lead to insufficient documentation, while too much documentation may lead to insufficient time for development. To mitigate this, I have included time for documentation in my timeline on a weekly basis. I will also integrate documentation into the development process to ensure consistency and accuracy.

3.4 Data protection

The project may involve handling or storing personal data, therefore compliance to data protection regulations and ensuring data is stored securely is mandatory

3.5 Copyright

The use of assets such as images, sounds, or certain tools without proper permission could lead to a copyright issue. To mitigate this, I will either create my own visuals or only use open-source/properly licensed materials and maintain a record of all external assets so I can cite them in my documentation. Additionally, research on pre-existing games on cybersecurity is imperative to avoid repeating the same game, which could also lead to copyright infringement.

References

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